ModelIt!

Teacher's Guide

Why Can We See Through Some Things?

The Science of Light and Materials

Quick Reference

Grade Level	Grade 1
NGSS Standard	1-PS4-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 1-PS4-3

Plan and conduct investigations to determine the effect of placing objects made with different materials in the path of a beam of light.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students investigate how different materials affect the amount of light that passes through them.
Disciplinary Core Idea	**PS4.B Electromagnetic Radiation** — Some materials allow light to pass through them, others allow only some light through, and others block all the light.
Crosscutting Concept	**Cause and Effect** — Students explore how the type of material causes different amounts of light to pass through.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Light is a form of energy that travels in straight lines. When light hits different materials, different things happen — and that is why we can see through some things but not others.

Three Kinds of Materials

Scientists sort materials into three groups based on how they interact with light. Transparent materials like clear glass and plastic wrap let almost all light pass through — that is why windows are made of glass! Opaque materials like wood, metal, and cardboard block all the light — you cannot see through a door. Translucent materials like frosted glass, wax paper, and thin fabric let some light through but scatter it, making things look blurry.

Why It Matters

Understanding how light passes through materials helps people make important choices every day. Builders choose glass for windows so sunlight can brighten rooms. Sunglasses are made with special translucent lenses that block some light to protect our eyes. Privacy screens on bathrooms use frosted glass that lets in light but keeps things private. Even the clothes we wear are chosen partly based on how much light they let through!

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify material type as something we choose (external) and light passing through as something that changes depending on the material (internal).

E **Establish Relationships**
Students figure out that transparent materials let the most light through, opaque materials block all light, and translucent materials are in between.

V **Visualize & Model**
Students build a simple model showing how the type of material affects how much light passes through.

E **Evaluate Outcomes**
Students test their model by shining a flashlight through different materials and recording what they observe.

R **Revise & Extend**
Students reflect on how their model helps explain everyday things like why windows are glass and walls are wood.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look around the room. Can you find something you can see through? Can you find something you cannot see through? Today we find out WHY!"*

Watch For

Have students point to things in the room that are see-through and not see-through. Make a quick list.

Slide 2: What Will We Learn?

3 min

Say: *"We are going to learn three big words today: transparent, translucent, and opaque. These words tell us how much light goes through things."*

Watch For

Teach the three words with hand motions: transparent = hands open wide, translucent = hands halfway, opaque = hands blocking.

Slide 3: Let's Be Scientists!

3 min

Say: *"Why can you see through a window but not through a wall? What do you think is different about them?"*

Watch For

Think-pair-share. Students whisper their idea to a partner, then share with the class.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts: Material Type is what the thing is made of. Light Passing Through is how much light gets to the other side."*

Watch For

Show three materials: clear plastic, wax paper, cardboard. Ask: Which one do you think lets the most light through?

Slide 5: How Are They Connected?

5 min

Say: *"Different materials do different things to light. Let us sort them! Transparent lets all light through, opaque blocks it, translucent lets some through."*

Watch For

Students sort material samples into three groups. Use sentence frame: This material is _____ because light _____.

Slide 6: What Did We Find Out? 5 min

Say: *"We discovered that the type of material decides how much light gets through. That is why windows are made of glass!"*

Watch For

Review discoveries. Shine a flashlight through each material type one more time to reinforce.

Slide 7: STEM Challenge 10 min

Say: *"Now you get to build a flashlight viewer! You will put different materials on the end and see how your view changes."*

Watch For

Distribute paper towel tubes, plastic wrap, wax paper, cardboard circles, tape, and flashlights. Students build and test.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Material Type
- **INTERNAL:** Light Passing Through

Relationships

- **Material Type to Light Passing Through:** POSITIVE (+) — More transparent materials let more light pass through. When the material is clearer, more light gets to the other side.

Key Discoveries

- Transparent materials let almost all light pass through so you can see clearly
- Opaque materials block all the light so you cannot see through them
- Translucent materials let some light through but things look blurry
- The type of material decides how much light gets through

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Light goes through everything a little bit"	Some materials block ALL light completely. These are called opaque. No light at all passes through a thick wall or a piece of cardboard, which is why you cannot see through them.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Clear things have holes in them that light goes through"	Transparent materials do not have tiny holes. The material itself lets light energy pass through it. Glass is completely solid with no holes, but light can still travel through it because of how the material is made.